DAT400 Project Proposal

Analyzing Inventory Metrics and Stock Returns Across Industries

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Executive Summary

Provide a brief overview of your project in 3-5 sentences. What will you build? What data will you use? What is the end goal? (75-150 words)

This project will analyze inventory metrics (inventory turnover, days in inventory, day sales in inventory, and inventory-to-sales ratio) across different industries and explore its relationship with stock returns. The data for this project will be inventory levels, cost of goods sold, industry classification, and historical stock returns for publicly traded companies in the S&P 500. Unless the data is already calculated, I will need to calculate the inventory turnover ratios and stock returns for each company. Using all this data, I will create an interactive dashboard that identifies trends, highlights how each sector performs, how companies compare within industry, and visualizes inventory metrics alongside stock performance.

Motivation

Why is this project important or interesting? Who would benefit from it? Include any relevant background information or statistics that support the value of your project. Cite your sources using numbered references, e.g., [1]. (100-200 words)

This project could benefit a range of people. Inventory turnover is one of the main metrics that businesses use to measure efficiency, cashflow, storage costs, and profitability [1][3]. Understanding which companies and industries have good inventory turnover could help investors make educated decisions. This project will also provide investors with insight into the relationship between stock return and inventory turnover. Inventory turnover rates vary by industry but are generally similar within industries [3]. Therefore, this project could prompt managers and executives who are at companies that compare worse to others within the same industry to investigate why their company is performing worse compared to others in the same industry. Furthermore, financial analysts, accountants, and college students with relevant majors could look at this project to gain deeper understanding of inventory turnover, stock returns, and overall financial performance across industries. Specifically for students, this project demonstrates how concepts they learn in class apply to real-world scenarios.

Data Question

State the main question your project will answer. This should be specific and measurable. (1-2 sentences)

How do inventory turnover and related efficiency metrics vary across industries and among companies within those industries? What is the relationship between these inventory metrics and stock returns?

Minimum Viable Product

List the concrete deliverables for your project. What will you produce? Be specific about: (50-100 words)
- *Visualizations or dashboards*
- *Models or algorithms*
- *Analysis or insights*

The primary deliverable for this project is a Tableau dashboard that will consist of many visualizations. First, bar charts, heatmaps, and boxplots will show inventory metrics by industry and company. Also, line charts will display how these metrics change over time by industry and company. Additionally, scatterplots will illustrate the relationship between inventory metrics and stock returns by industry and company. In the presentation showing off this dashboard, there will be accompanying analysis explaining trends, patterns, comparisons and correlation from these visualizations.

Tools and Technologies

List the tools, languages, and libraries you plan to use. (e.g., Python, pandas, scikit-learn, Streamlit, PostgreSQL, etc.)

- Python
- Jupyter Notebooks
- Pandas
- NumPy
- Seaborn
- PostgreSQL
- Docker
- Datagrip
- Tableau
- Git
- Github

Schedule

Fill in your planned milestones with target dates.

Milestone	**Target Date**
Data acquisition	4/2
Data cleaning	4/10
Model/analysis development	4/24

Presentation preparation	4/30
Demo day	5/1

Data Sources

List all data sources you plan to use.

- SEC EDGAR API
- Yahoo Finance Library
- Kaggle Dataset: S&P 500 Companies List with Sectors

Related Work (Optional)

Have others attempted similar projects? What existing tools or research is relevant? How will your project differ or build upon prior work? (50-100 words)

Similar research has previously been done. For example, one study investigated inventory turnover by mode of operation [2]. The proposed project differs from this one by investigating inventory turnover across companies and industries, rather than the modes of operation.

Known Issues and Challenges

Identify potential challenges you anticipate and how you plan to address them. Be honest about areas where you may need additional help or research. (75-150 words)

There are several challenges that I may encounter throughout this project. The first challenge is domain knowledge. I have a limited background dealing with financial data, which could affect my decisions during data cleaning or when interpreting the data or visualizations. To avoid this, I will research the inventory metrics, stock returns, and any unfamiliar concepts. Another challenge involves the data itself. I have not yet explored the data, so there is a good chance that I will run into problems when cleaning it, especially when combining the data from various sources. If I run into technical problems, I will find online sources, such as tutorials or documentation, to help me work out a solution. While these are the only challenges I can identify at this moment, I anticipate there will still be unexpected challenges, and I will combat them as best I can when they arise.

References

List any sources cited in your proposal.

1. Jason Fernando, “Inventory Turnover Ratio: What It Is, How It Works, and Formula,” Investopedia, September 16, 2024, https://www.investopedia.com/terms/i/inventoryturnover.asp.

2. Junaid Ali Khan and Shiming Deng, “Analyzing Inventory Turnover Performance in Different Retailing Mode of Operations,” European Scientific Journal 12, no. 34 (2016): 34–45, https://www.researchgate.net/publication/312100223_Analyzing_Inventory_Turnover_Performance_in_Different_Retailing_Mode_of_Operations.
3. Onramp Funds, “Inventory Turnover Benchmarks by Industry 2025,” Onrampfunds.com, October 10, 2025, https://www.onrampfunds.com/resources/inventory-turnover-benchmarks-by-industry-2025.